



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
28-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TAAS, TTAS & TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A particle of specific charge ' α ' is projected from origin at $t = 0$ with a velocity $\vec{V} = V_0 (\hat{i} + \hat{k})$ in a magnetic field $\vec{B} = -B_0 \hat{k}$. Then :

(A) at $t = \frac{\pi}{\alpha B_0}$, velocity of the particle is $-V_0 (\hat{i} - \hat{k})$

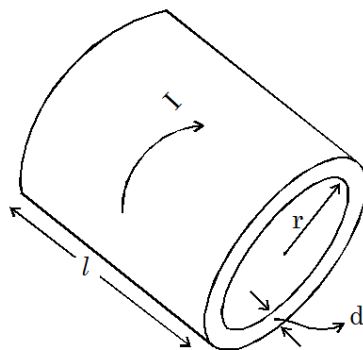
(B) at $t = \frac{\pi}{4\alpha B_0}$, speed of the particle is V_0

(C) at $t = \frac{2\pi}{\alpha B_0}$, magnitude of displacement of the particle is more than $\frac{2V_0}{\alpha B_0}$

(D) at $t = \frac{2\pi}{\alpha B_0}$, distance travelled by the particle is less than $\frac{2\sqrt{2}\pi V_0}{\alpha B_0}$.

Space for Rough Work

2. A hollow cylinder has length ℓ , radius r , and thickness d where $\ell \gg r \gg d$, and is made of a material with resistivity ρ . Initially a current I_0 is set up in cylinder and now source is removed so a time varying current I flows through the cylinder in tangential direction. Assume the current is always uniformly distributed along the length of cylinder i.e. $\frac{dI}{d\ell}$ is constant for entire cylinder. The cylinder is fixed so that it can not move. Assume that there are no externally generated magnetic field during the time considered for this problem. Choose correct statement(s) :-



- (A) Magnetic field B inside the cylinder when current through cylinder I is given by $B = \frac{\mu_0 I}{\ell}$
- (B) When current is changing at a rate of $\frac{dI}{dt}$, then emf e developed along the circumference of cylinder is
- $$e = -\frac{\pi \mu_0 r^2}{\ell} \frac{dI}{dt}$$
- (C) Relation between induced emf e and current I is given by $e = I \rho \cdot \frac{2\pi r}{\ell d}$
- (D) At time $t = 0$, if current in cylinder is I_0 , the current I at any time t is given by $I = I_0 e^{-\alpha t}$ where $\alpha = \frac{2\rho}{\mu_0 r d}$

Space for Rough Work

3. The figure-2, figure-3, figure-4 and figure-5 represents the graphs of the source voltage V , resistor voltage V_R , inductor voltage V_L and capacitor voltage V_C as a function of time t of the series circuit with impedance 500Ω as shown in the figure-1 ($t_0 = \frac{\pi}{20}\text{ms}$). Choose correct option(s).

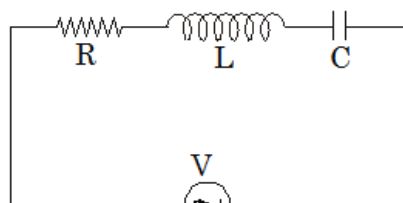


Figure-1

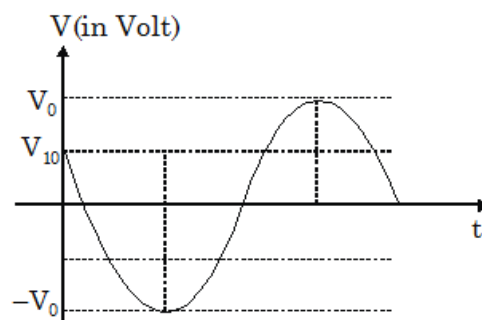


Figure-2

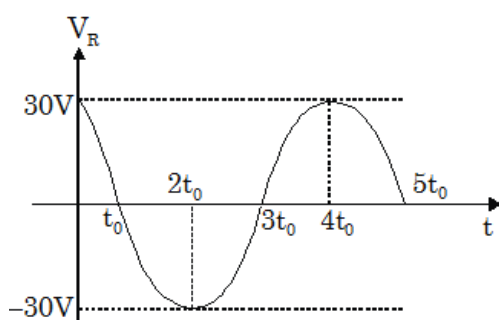


Figure-3

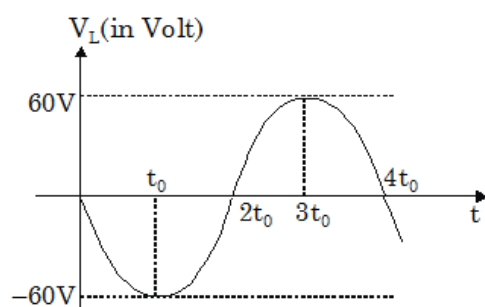


Figure-4

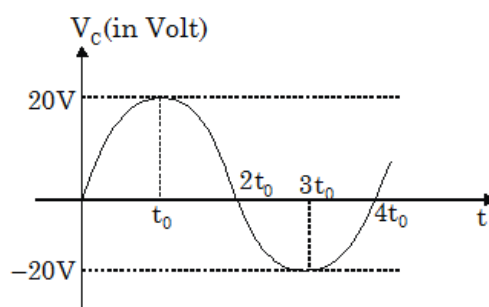
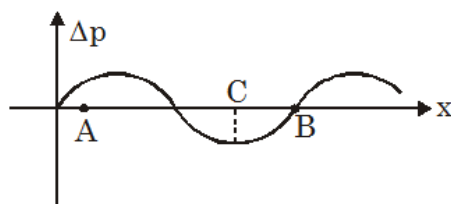


Figure-5

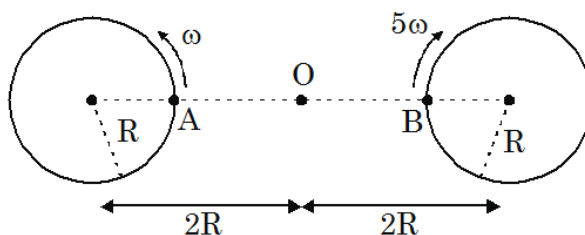
- (A) $R = 400\Omega$, $C = 0.5\mu\text{F}$ and $L = 50\text{mH}$
 (B) $R = 300\Omega$, $C = 0.5\mu\text{F}$ and $L = 60\text{mH}$
 (C) $V_{10} = 30\text{V}$
 (D) $I = 0.1\text{A} \cos(10000t)$ Amp and current lag the source voltage

Space for Rough Work

4. The variation of pressure for a sound wave travelling along the positive x axis is shown below. Choose the correct statements :-



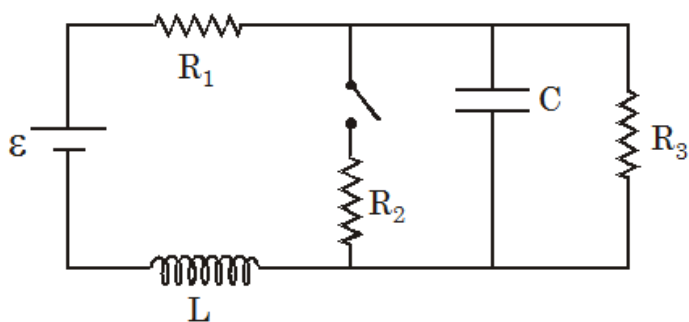
- (A) Particles just on left and right of point A are moving in opposite direction.
 (B) Speed of particle on just left of A is less than that on right of A.
 (C) Particles on just left and just right of B have opposite direction of motion.
 (D) Particles on just left and just right of C are moving in same direction.
5. Two sound source A and B start moving on circular path of radius R with angular speed ω and 5ω respectively. An observer is at position 'O' on the line joining the centers. At $t = 0$ position of sources are represented in the figure. Frequency of source A and B is 120 Hz each. [Speed of sound is 320 m/s, $R = 20$ m, $\omega = 1/5$ rad/s]



- (A) Maximum beat frequency observed by observer is 7.48 Hz
 (B) Maximum beat frequency observed by observer is 9.48 Hz
 (C) Minimum beat frequency observed by observer is 6.28 Hz
 (D) Minimum beat frequency observed by observer is 3.14 Hz

Space for Rough Work

6. The circuit shown in the diagram is in steady state with the switch open. When the switch is closed, which of the following will not change immediately?



- (A) Potential difference across capacitor C . (B) Current through the inductor L .
 (C) Potential difference across resistance R_1 . (D) Current through the resistance R_2 .

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

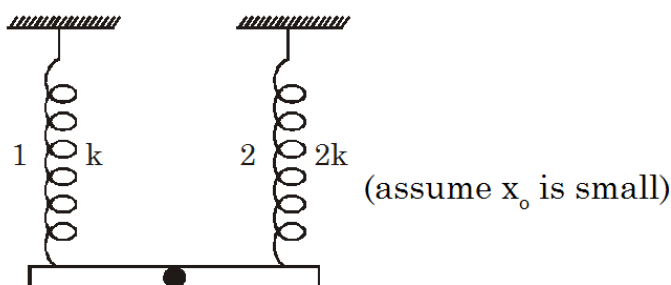
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Two light vertical springs 1 and 2 of stiffness K and $2K$ are connected to a light horizontal rod at its two ends as shown. A bead of chewing gum of mass ' m ' is stuck gently at the mid point of the rod and lowered slowly, such that where equilibrium is reached the bead displaced by x_0 .

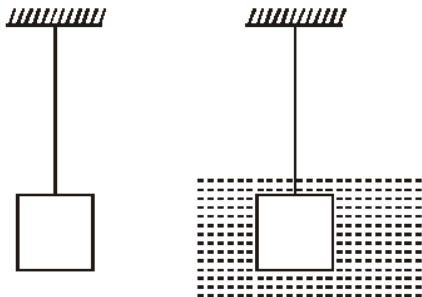


7. The energy of oscillation of the bead for an amplitude y_0 is equal to
- (A) mgx_0 (B) $mg(x_0 + y_0)$ (C) $\frac{3}{2}ky_0^2$ (D) $\frac{4}{3}Ky_0^2$
8. The period of SHM of the bead is
- (A) $2\pi\sqrt{\frac{3m}{8K}}$ (B) $2\pi\sqrt{\frac{m}{3K}}$ (C) $3\pi\sqrt{\frac{2m}{K}}$ (D) None of these

Space for Rough Work

Paragraph for Questions 9 and 10

A cylindrical block of density 2 gm/cc is suspended by a thin wire 3m long. When the wire is vibrated, it produces fundamental frequency of 280 Hz . The mass of wire is so small that tension in the wire can be assumed to be constant. Now the block is dipped in a liquid. It is found that the first overtone now is 280 Hz .



9. What is the density of the liquid?
- (A) 1 gm/cc (B) 1.5 gm/cc (C) 0.75 gm/cc (D) 1.25 gm/cc
10. If the block is dipped to $1/3$ of its height in the liquid. What is the new fundamental frequency?
- (A) 140 Hz (B) 210 Hz (C) $140\sqrt{3} \text{ Hz}$ (D) $140\sqrt{2} \text{ Hz}$

Space for Rough Work

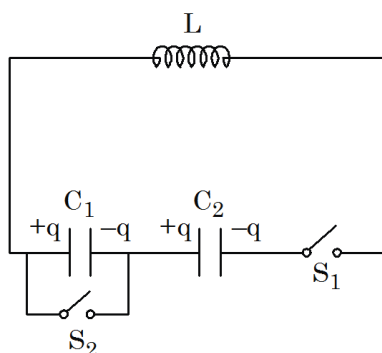
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

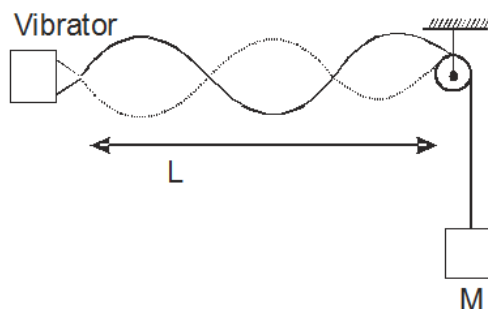
1. An alternating voltage of 260 volt and $\omega = 100$ radian/second, is applied in an LCR series circuit where $L = 0.01$ H, $C = 4 \times 10^{-4}$ F and $R = 10\Omega$. The power supplied by the source is 200k. Find the value of k.
2. Two capacitors $C_1 = C_2 = \frac{1}{\pi^2} \times 10^{-2}$ F and inductor $L = 2 \times 10^{-2}$ H are connected in series as shown in the figure. Initially charge on each capacitors are $4\sqrt{3} \mu\text{C}$. At $t = 0$ switch S_1 is closed and $t = \frac{1}{400}$ sec, switch S_2 is also closed. The maximum charge on capacitor C_2 during LC oscillation is $k\sqrt{2} \mu\text{C}$. Find k.



3. The coil in a moving coil Galvanometer experiences torque proportional to current passed through it. If a steady current i is passed through it the deflection of the coil is found to be 90° . Now the steady current is switched off and a charge Q is suddenly passed through the coil. If the coil has N turns of area A each and its moment of inertia is I about the axis it is going to rotate and placed in uniform magnetic field B then the maximum angle through which it deflects upon passing Q is $Q\sqrt{\frac{\pi NAB}{kiI}}$. Find the value of k. (Assume damping is small and it takes large number of oscillations for coil to eventually stop)

Space for Rough Work

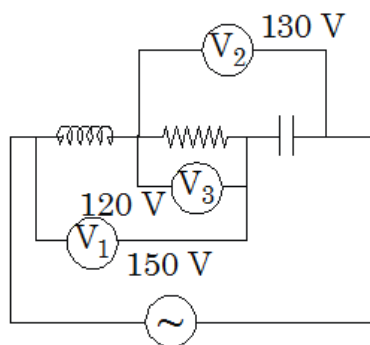
4. The mechanical energy between a given node and other node on a string is $\frac{4\rho A^2 \omega^2 \pi S}{k}$, where ρ is the mass density of the string, S is the cross-sectional area, A is the amplitude of two harmonic waves of angular velocity ω travelling on the string in opposite directions. Find the number of loops for which mechanical energy is given as above
5. In the given sonometer arrangement, a mass M can be hung from a string, that passes over a light pulley. The string connected to a vibrator having constant frequency. When the value of M is either 16 Kg or 25 Kg standing waves are observed, however, no standing waves are observed with any mass between these values. The largest mass for which standing waves could be observed is $n \times 10^2$ Kg and frequency of vibrator is f (Hz) then find the numerical value of $n \times f$. (mass per unit length for string is 10^{-3} kg/m and length $L = 2$ m)



6. An observer at distance r from point source observes intensity at decibal scale β dB. Now eight coherent identical point source [all are in same phase] are placed at the position of previous source and observer move to a distance $3r$ from sources. Now observed sound intensity is $(\beta + n)$ dB. Find the value of n . ($\log_{10} 2 = 0.3$; $\log_{10} 3 = 0.47$)

Space for Rough Work

7. A charged particle sits at the centre of a circular region containing a magnetic field, $\vec{B}$. The field $\vec{B} = \vec{B}(r)$ depends only on radial position r , and is perpendicular to the plane of circle. The total magnetic flux through the circle is zero. The particle is given a kick so that it leaves the circular region, then at the instant it leaves, its velocity makes an angle $n \times \frac{\pi}{4}$ with outward radial direction. Find n .
8. The rms readings of voltmeters V_1 , V_2 , V_3 are 150V, 130V and 120V respectively. If applied ac rms voltage = x and power factor of the circuit is y . Find $\frac{x \times y}{20}$.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A galvanic cell is composed of two hydrogen electrodes, one of which is a standard hydrogen electrode. Identify correct option(s) if other electrode is immersed in :

Use : $\frac{2.303RT}{F} = 0.06$, $T = 298 \text{ K}$

(I) 0.1 M HCl (aq.) ,

(II) 0.1 M CH_3COOH (aq.) [$K_a = 10^{-5}$],

(III) 0.1 M NaOH (aq.)

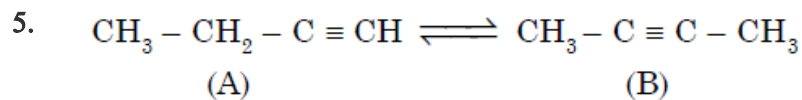
(IV) 0.1 M NH_4OH (aq.) [$K_b = 10^{-5}$].

- (A) For case III cell EMF is of maximum magnitude among given options
(B) For case - IV : $E_{\text{cell}} = 0.66$ volts
(C) For case - I : cell EMF is of minimum magnitude among given option
(D) For case - II : $E_{\text{cell}} = 0.18$ volts

Space for Rough Work

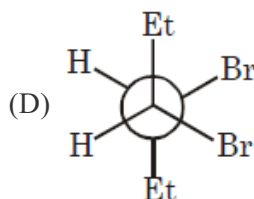
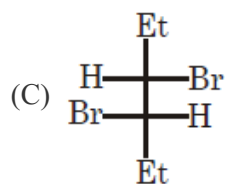
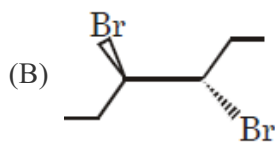
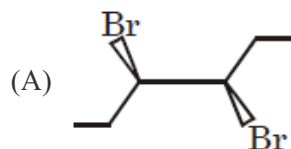
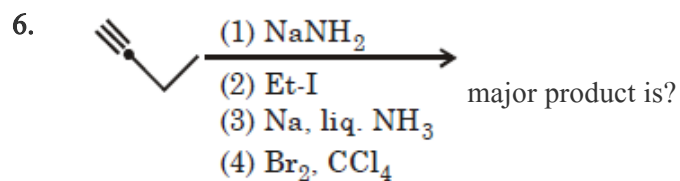
2. Select the correct statement -
- (A) In a mixture of NH_4Cl & NH_4OH , when further NH_4Cl is added, pH of mixture decreases.
- (B) Generally there is negligible effect of dilution on pH of buffer solution
- (C) Titration of NaHCO_3 with HCl can be indicated in presence of phenolphthalein indicator.
- (D) There is no sudden change in pH near equivalence point in weak acid-weak base titration.
3. Which of the following is/are ore of Fe ?
- (A) Magnetite (B) Magnesite (C) Limonite (D) Siderite
4. Which of the following option is/are involved in copper extraction ?
- (A) FeO impurity is removed as FeSiO_3
- (B) In bessemerisation blister copper is formed
- (C) Poling is used for purification of copper metal
- (D) In electrolytic refining of copper metal Zn rod used as anode and copper sheet used as cathode

Space for Rough Work



Which option is correct regarding this interconversion ?

- (A) To carry out forward reaction alcoholic KOH / Δ is required
 (B) To carry out backward reaction alcoholic KOH / Δ is required.
 (C) A is more acidic than B
 (D) In the intermediate ($\text{CH}_2 = \text{C} = \text{CH} - \text{CH}_3$) formed during this interconversion, maximum number of atoms present in same plane are 5.



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

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- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Solution of a weak acid and its anion (that is, its conjugate base) or of a weak base and its common cation are buffer solution. When we add a small amount of acid or base to any one of them, the pH of solution changes very little. pH of buffer solution can be computed as

$$\text{for acidic buffer : } \text{pH} = \text{pK}_a + \log \frac{[\text{Conjugate base}]}{[\text{Acid}]}$$

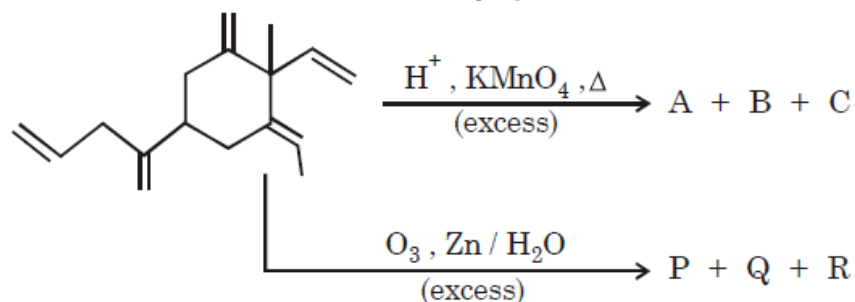
$$\text{for basic buffer : } \text{pOH} = \text{pK}_b + \log \frac{[\text{Conjugate acid}]}{[\text{Base}]}$$

It is generally accepted that a solution has useful buffer capacity (pH change resisting power) provided that the value of [salt or conjugate base]/[acid] for acidic buffer lies within the range of 1 : 10 to 10 : 1. Buffer capacity is maximum, when [Conjugate base] = [acid]

7. Calculate the pH of a solution made by adding 0.01 mole of HCl in 100 mL of a solution which is 0.2 M in NH_3 ($\text{pK}_b = 4.74$) and 0.3 M in NH_4^+ : ($\log 2 = 0.3$)
- (A) 5.34 (B) 8.66 (C) 7.46 (D) 9.26
8. Select correct statement:
- (A) When we add small amount of NaOH in acidic buffer solution, pOH of solution is increases
- (B) When we add small amount of NaOH in basic buffer solution, pH of solution is increases
- (C) When we add small amount of water in acidic buffer solution, pH of solution is decreases
- (D) When 100 mL of 0.2 M CH_3COOH react with 200 mL of 0.1 M NaOH, buffer solution is formed

Space for Rough Work

Paragraph for Questions 9 and 10

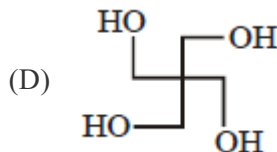
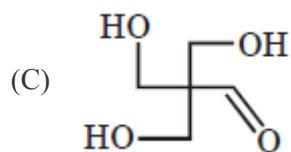
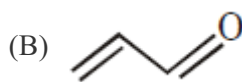


A, B, C, P, Q and R all are carbon containing compounds out of which A and P are cyclic compounds:

9. Number of carbon in one molecule of A is :

- (A) 8 (B) 9 (C) 10 (D) 11

10. What will be the product if Q and R are treated with KOH in same proportion in which they are formed in this reaction ?



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Equivalent conductivity of AgNO_3 , CaCl_2 and $\text{Ca}(\text{NO}_3)_2$ at infinite dilution are 110, 230 and $196.5 \Omega^{-1} \text{cm}^2 \text{eqv}^{-1}$ respectively. Conductivity of $\text{AgCl}(\text{aq.})$ & H_2O is $2.4 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$ and $1.4 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$ respectively. Find solubility of AgCl in mg/litre. [Atomic mass : $\text{Ag} = 108$, $\text{Cl} = 35.5$]
2. 0.001 mole of Potassium acid oxalate $\text{K}_2\text{C}_2\text{O}_4 \cdot 3\text{H}_2\text{C}_2\text{O}_4 \cdot 4\text{H}_2\text{O}$ can be oxidized by MnO_4^- in acid medium. Calculate the volume (in ml) of 0.2 M KMnO_4 needed in acid solution.
3. What is the sum of oxidation number and coordination number of central metal atom of complex formed as yellow precipitate when CoCl_2 reacts with excess of potassium nitrite?
4. Total number of ore which can be concentrated by magnetic separation method.
Tin stone consisting of Wolframite, Ilmenite, Chromite, Silver glance, Dolomite
5. $\text{Salt 'X'} \xrightarrow{\text{dil. H}_2\text{SO}_4} \text{No observation}$
 $\text{Salt 'X'} \xrightarrow[\Delta]{\text{MnO}_2 + \text{conc. H}_2\text{SO}_4} \text{Coloured gas/vapours}$

Total number of anions which will give coloured gas/vapour will be :-

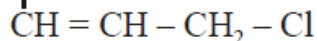


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6. During extraction of iron, CO is the chief reducing agent. If 72 Kg of carbon is available and produced CO is completely used to reduce Haematite (present in excess amount) then calculate how much Kg of Iron is obtained after reduction.

(Given : atomic weight of C-12 g/mol, O-16 g/mol, Fe-55.85 g/mol)

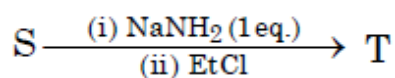
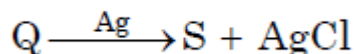
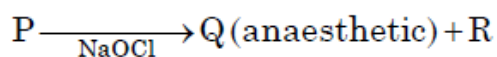
7. $\text{H}_3\text{C}-\text{C}=\text{O}$



How many reactions are possible for the above compound?

- (i) Nucleophilic substitution (ii) Free radical substitution
(iii) Electrophilic addition (iv) Nucleophilic addition
(v) Aldol condensation (vi) Electrophilic substitution

8. $\text{CH}_3-\text{CH}_2-\text{C}\equiv\text{CH} \xrightarrow{\text{H}_3\text{O}^+/\text{HgSO}_4} \text{P}$
(X)



How many statements are correct?

- (1) P gives positive test with NaHSO_3
(2) Q on aerial oxidation gives poisonous gas
(3) S is formed when CaC_2 undergoes hydrolysis.
(4) X and T are identical compound

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f(x)$ be a non-negative continuous and bounded function for all $x \geq 0$.
If $(\cos x) f'(x) \leq (\sin x - \cos x) f(x) ; \forall x \geq 0$, then which of the following is / are correct ?

(A) $f(6) + f(5) > 0$

(B) $x^2 - 3x + 2 + f(7) = 0$ has 2 distinct solutions

(C) $f(5)f(7) - f(6)f(5) = 0$

(D) $\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x - 6)} = 1$

2. Solution of differential equation $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \cdot \sin y$ is
(where c is the constant of integration)

(A) $2y = \sin y (1 - cx^2)$

(B) $2x = \cot y (1 + 2cx^2)$

(C) $2x = \sin y (1 + 2cx^2)$

(D) $2x \sin y = 1 - 2cx^2$

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3. $\int \left(2e^{\tan^{-1}x + x^2 + \ln x} + \frac{(2x^3 + 2x + 1)}{\left(x + \frac{1}{x}\right)} e^{\tan^{-1}x + x^2 + \ln x} \right) dx$ is equal to

(where C is constant of integration)

(A) $\frac{x^3}{3} e^{\tan^{-1}x + x^2 + \ln x} + C$

(B) $\frac{x^2}{2} e^{\tan^{-1}x + x^2 + \ln x} + C$

(C) $x e^{\tan^{-1}x + x^3 + \ln x} + C$

(D) $x^2 e^{\tan^{-1}x + x^2} + C$

4. Let $f(x)$ be the function defined by

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} \frac{\left(1 + \left[\frac{\sin x}{x}\right]^{2n}\right) \{x\}}{1 + \left[\frac{x}{\tan x}\right]^{2n}} & ; 0 < x < \frac{\pi}{2} \\ \frac{\{x\}^3 - 3\{x\}}{4 - [x]} & ; -2 < x \leq 0 \\ x^3 - 27x - 46 & ; x \leq -2 \end{cases}$$

(where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part function) then which of the following options is/are true

(A) there exist exactly 2 points of local minima of $f(x)$ in the interval $\left(-4, \frac{\pi}{2}\right)$

(B) The sum of abscissa of all of its point of local maxima is -4

(C) The sum of abscissa of all of its point of local minima is 1

(D) $f(x)$ is not differentiable in the interval $(-\infty, -1)$

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5. A polynomial $f(x)$ of the least degree, has a local maximum value equal to 6 at $x = 1$, and a local minimum value equal to 2 at $x = 3$, then
(where $[.]$ represent greatest integer function)
- (A) The equation $f(x) = 3$, has three distinct roots.
- (B) $f(x)$ is strictly decreasing in $1 < x < 2$
- (C) $f(0) = 2$
- (D) If α is real root of $f(x) = 0$, then $[\alpha] = -1$
6. If $f(1) = e^2$ and $\frac{d}{dx}(f(x)) > 3x^2 f(x) \forall x > 1$ then $f(x)$ cannot take values (where $e = 2.71$)
- (A) e (B) $\frac{9}{2}$ (C) 8 (D) 10

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SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(a) = b$ and $f'(a) = f''(a) = 0$. Suppose $g(x) = f(x) - f(a) + (a - x)f'(x) + 3(x - a)^2$ for which conclusion of Rolle's theorem in $[a, b]$ holds at $x = 2$, where $2 \in (a, b)$

7. The value of $f''(2)$, is
- (A) 2 (B) 4 (C) 0 (D) 6
8. If the value of definite integral $\int_a^b f(x) dx$ is $\frac{\lambda}{\mu}$ (where λ & μ are relatively prime numbers), then $(\lambda - 5\mu)$ equals:
- (A) 1 (B) 2 (C) 3 (D) 0

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Paragraph for Questions 9 and 10

A function $f(x)$ having the following properties:

- (i) $f(x)$ is continuous except at $x = 3$
- (ii) $f(x)$ is differentiable except at $x = -2$ and $x = 3$
- (iii) $f(0) = 0$, $\lim_{x \rightarrow 3} f(x) \rightarrow -\infty$, $\lim_{x \rightarrow -\infty} f(x) = 3$, $\lim_{x \rightarrow \infty} f(x) = 0$
- (iv) $f'(x) > 0 \forall x \in (-\infty, -2) \cup (3, \infty)$ and $f'(x) \leq 0 \forall x \in (-2, 3)$
- (v) $f''(x) > 0 \forall x \in (-\infty, -2) \cup (-2, 0)$ and $f''(x) < 0 \forall x \in (0, 3) \cup (3, \infty)$

9. The maximum possible number of solutions of $f(x) = |x|$ is(are)

- (A) 1 (B) 2 (C) 3 (D) 0

10. If $f'(0) = 0$, then number of points where the graph of function $y = f(-|x|)$ is non-differentiable is

- (A) 2 (B) 3 (C) 4 (D) 0

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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\lambda = \int_1^{\infty} \frac{1}{x^2} \ln \left(\frac{x^2}{x-1} \right) dx$ then $\frac{\lambda}{4}$ equals
2. Let $g : \mathbb{R} \rightarrow \{4\}$ be a function given by $g(x) = x^3 (f'(t) - 2) + x^2 f''(t) + 4x (f(0) + 6) + 4, \forall t \in \mathbb{R}$
and $h(x)$ is defined as $h(x) = \begin{cases} \int_0^x |f(t) - 2| dt, & 0 \leq x \leq 6 \\ (x-6)^2 + 20, & 6 < x \leq 12 \end{cases}$. If number of integers in the range of $h(x)$ is N then, the value of $\frac{N}{38}$ equals
3. If $y = mx + c$ is bitangent (a line is tangent to the curve at two distinct points) on the curve $y = x^4 + 2x^3 - 11x^2 - 13x + 35$, then the value of $\frac{|m+c|}{5}$ is
4. Let θ be the acute angle between the curves, $y = x^x \ln x$ and $y = \frac{2^x - 2}{\ln 2}$ at their point of intersection on the line $y = 0$. The value of $\tan \theta$ is equal to :-

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5. Let $I_n = \int_0^{\frac{3\pi}{2}} (\ln |\sin x|) \cos(2nx) dx$ where $n \in \mathbb{N}$. If $(12I_3 - 16I_2)$ is $k\pi$ then value of $\frac{k}{4}$ equals
6. If $\lim_{n \rightarrow \infty} \left[\left((2017)^{2018} \right)^n + \left((2018)^{2017} \right)^n \right]^{1/n}$ is equal to a^b where $a, b \in \mathbb{N}$ then $\left(\frac{b-a}{4} \right)$ is equal to
7. If n is even, then the number of real roots of the equation $1 + 2x + 3x^2 + \dots + (n+1)x^n = 0$ is
8. Let $y = f(x)$ be a curve C_1 passing through $(2,2)$ and $\left(8, \frac{1}{2}\right)$ and satisfying the differential equation $y \left(\frac{d^2y}{dx^2} \right) = 2 \left(\frac{dy}{dx} \right)^2$. If the shortest distance between C_1 and C_2 is $(\sqrt{a} - b)$ (Curve C_2 is $x^2 + y^2 = 4$) where $a, b \in \mathbb{N}$, then $\frac{(a+b)}{4}$ is equal to

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator**Space for Rough Work**